VISIONATTEND PRO

AI-Powered Classroom Attendance & Biometric Analytics Platform

Official Platform Title:	VisionAttend Pro - AI Classroom Attendance System
Document Classification:	Final Technical Project Specification & Verified System Dossier
Release Version:	v1.0.0 (Production Architecture)
Core Vision Architecture:	YOLOv8-Face Multi-Face Detection + ArcFace ResNet-50 512-D ONNX Embeddings
Matching Engine:	Normalized Cosine Similarity (Default Tau = 0.50) + Ambiguity Margin Guard
Backend Stack:	FastAPI 0.109.0 (Python 3.10.11) + SQLAlchemy 2.0.52 + SQLite 3
Security & Auth:	JWT HS256 (PyJWT 2.13.0) + Bcrypt 5.0.0 + Role-Based Access Control
Date of Release:	August 30, 2026
Verification Status:	**24/24 Automated Regression Tests Passed (100%)**

Executive Summary: VisionAttend Pro automates university classroom attendance using multi-camera panoramic group photos (1 to 4 angles) or live webcam feeds. Eliminating traditional roll calls and proxy attendance, the system detects multiple students in seconds, extracts 512-dimensional facial embeddings, matches identities against enrolled course rosters, and compiles regulatory Excel spreadsheets and signed PDF compliance dossiers.

1. Abstract

VisionAttend Pro is an enterprise-grade automated classroom attendance and biometric analytics platform designed to eliminate manual roll calls and proxy sign-ins in academic institutions. Operating on a dual-stage computer vision architecture, the platform pairs **YOLOv8-Face** (yolov8n-face.pt) for high-density multi-face localization with **ArcFace ResNet-50 ONNX** (arcface_w600k_r50.onnx) for 512-dimensional angular hypersphere feature extraction. Attendance is recorded from multi-perspective classroom photos (1–4 angles) or real-time webcam feeds. Evaluated student embeddings are matched against course rosters using Normalized Cosine Similarity with Ambiguity Margin Protection. Enrolled students recognized above threshold (default $\tau = 0.50$) are recorded as PRESENT, while unobserved roster members are marked ABSENT. Unrecognized faces route to an Unidentified Face Resolution Queue. Built upon FastAPI, SQLAlchemy, SQLite, and an ES6 modular Single Page Application, VisionAttend generates multi-sheet division-wise Excel workbooks (.xlsx) and PDF dossiers (.pdf) tracking $<75\%$ attendance defaulters. The platform is verified by a 24-point regression suite with 100% pass rate.

2. Problem Statement & Motivation

Conducting manual attendance in higher education consumes 10–15 minutes of lecture time for classes with 60 to 120 students. Paper sign-in sheets are susceptible to proxy attendance, while doorway fingerprint scanners cause physical bottlenecks. VisionAttend Pro provides a touchless, crowd-scale solution that photographs the classroom, deduplicates identities across angles, and synchronizes attendance records in seconds with zero hardware touchpoints.

3. System Architecture & Component Layers

The platform is structured into clean architectural tiers with strict separation of concerns:

- **Client Tier (Frontend SPA)**: Modular ES6 JavaScript view controllers, modular CSS architecture, Chart.js weekly analytics, and Lucide SVG iconography.
- **API Tier (FastAPI)**: Asynchronous REST route controllers for authentication, student registry, course catalogs, session ingestion, records, and reports.
- **AI / Computer Vision Core**: YOLOv8-Face multi-face detection, ArcFace ResNet-50 ONNX embedding extraction, FFT spectral anti-spoofing, and cosine similarity matching.
- **Persistence Tier**: SQLite relational database (database/attendance.db) managed via SQLAlchemy 2.0 ORM with automated migrations and backup archives.
- **Storage Tier**: Runtime storage for student reference photos, session photos, unknown face crops, and generated reports (data/).

4. Technology Stack & Verified Versions

Technology	Version	Layer	Functional Purpose
Python	3.10.11	Runtime	Core execution runtime
FastAPI	0.109.0	Backend	Async web API framework & OpenAPI docs
SQLAlchemy	2.0.52	ORM	Database ORM & schema management
SQLite	3.x	Database	Relational storage (database/attendance.db)
Ultralytics (YOLO)	8.4.134	Computer Vision	YOLOv8-Face multi-face detector
ONNX Runtime	1.23.2	AI Inference	ArcFace ResNet-50 512-D neural execution
PyTorch	2.10.0+cpu	Deep Learning	Tensor computation for YOLOv8
OpenCV (cv2)	4.13.0	Image Processing	Image decoding, resizing & bbox overlays
PyJWT	2.13.0	Security	JWT token encoding/decoding (HS256)
Bcrypt	5.0.0	Security	Password cryptographic salting (12 rounds)
Pydantic	2.13.5	Validation	Request/response schema validation
OpenPyXL	3.1.2	Reporting	Styled multi-sheet Excel (.xlsx) exports
ReportLab	5.0.1	Reporting	Automated compliance PDF report compilation
Pytest	9.0.3	Testing	Automated 24-point regression test suite

5. AI Pipeline: YOLOv8-Face & ArcFace ResNet-50

The facial recognition pipeline separates detection from identity representation:

1. **YOLOv8-Face Detection (Localization)**: Single-stage CNN model (yolov8n-face.pt) localizes all faces in classroom images down to 20×20 px, generating bounding boxes with confidence scores $c_{\text{det}} \ge 0.30$.
2. **Facial Preprocessing**: Face crops are extracted with 15% contextual margin, resized to 112×112 pixels, normalized to range $[-1, 1]$, and verified for spectral moire artifacts via 2D Fast Fourier Transform.
3. **ArcFace ResNet-50 Feature Representation**: ONNX engine computes a 512-dimensional continuous feature vector normalized to unit length on the hypersphere ($\lVert \mathbf{e} \rVert_2 = 1.0$).
4. **Normalized Cosine Similarity Matching**: Vector dot product $S_C = \mathbf{e}_q \cdot \mathbf{e}_s$ evaluates similarity against enrolled class roster vectors. Matches exceeding threshold $\tau=0.50$ with ambiguity margin $\Delta \ge 0.05$ are confirmed as present students; unmatched faces route to the Unknown Faces Resolution Queue.

6. REST API Architecture (39 Endpoints)

The FastAPI backend exposes 39 structured REST endpoints protected by JWT Bearer token authentication and Role-Based Access Control:

Method	Endpoint URL	Purpose / Action	Role Access
POST	/api/auth/login	OAuth2 form login token generator	Public
POST	/api/auth/login-json	JSON payload SPA login endpoint	Public
GET	/api/auth/me	Current authenticated user profile	All Auth
GET/POST	/api/auth/users	List or create institutional faculty	Admin Only
GET/POST	/api/classes	List or create academic courses	All Auth
POST	/api/classes/{id}/enroll	Enroll student roster into course	All Auth
GET	/api/students	Query students with dept/sem/div filters	All Auth
POST	/api/students/register-with-photo	Register student & extract 512-D vectors	All Auth
POST	/api/sessions/create-and-process	Ingest 1-4 photos & run YOLO+ArcFace	All Auth
GET	/api/sessions/{id}	Get session details & annotated photo URLs	All Auth
PUT	/api/attendance/records/{id}	Manual override of attendance status	All Auth
GET	/api/unknown-faces	Query unidentified face crops queue	All Auth
POST	/api/unknown-faces/{id}/tag	Tag unknown face to registered student	All Auth
GET	/api/reports/advanced-data	Aggregated attendance data table	All Auth
GET	/api/reports/export/excel	Download multi-sheet Excel (.xlsx)	All Auth
GET	/api/reports/export/pdf	Download signed compliance PDF dossier	All Auth
GET	/api/analytics/dashboard	Dashboard KPIs & trend curve metrics	All Auth
GET	/api/admin/system-diagnostics	System telemetry & database statistics	Admin Only
POST	/api/admin/backup-database	Generate timestamped SQLite backup	Admin Only

7. Database Telemetry & Live System Inventory

Audited directly from production storage (database/attendance.db):

Entity Domain	Verified Count	Operational Description
Users (Faculty / Admin)	4 Accounts	admin (Admin), dr_sharma, prof_anil, prof_sunita (Faculty)
Academic Courses / Classes	5 Courses	CS-301, AI-201, CS-302, 520 (MongoDB), 522 (GAN)
Registered Students	5 Students	All profiles enrolled with active 512-D ArcFace vectors
Attendance Sessions Logged	88 Sessions	Lecture sessions processed with multi-face detection
Attendance Records	259 Records	Individual biometric attendance entries logged
Unidentified Faces Queue	729 Face Crops	Cropped facial images preserved for audit/tagging
Automated Pytest Suite	24 / 24 Passed	100% test pass rate across auth, AI, DB, and reports

8. Verified Technical Conclusion

VisionAttend Pro successfully solves the problem of automated classroom attendance. By leveraging **YOLOv8-Face** for multi-face localization and **ArcFace ResNet-50 (512-D)** for angular hypersphere feature extraction, the platform achieves reliable identity verification without proxy loopholes. Operating on a clean, professional industry-standard structure, the system delivers institutional security with JWT/Bcrypt, multi-division Excel/PDF reports, automated SQLite snapshot backups, and an authenticated faculty workspace. All 24 automated tests pass with 100% success rate, proving the platform is production-ready.